

TDM — Trust Degradation Module

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Modul3 TDM — Trust Degradation Module

OriginID: OOF-OID-GOV-TDM-2026-06-02-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Operational Reality Standards™

Operational Layer: Autonomous Systems Governance Layer

Governed Space: Trust Degradation Governance

Category: Governance & Enforcement

Subcategory: Operational Trust Degradation

Type: Operational Risk & Trust Integrity Module

Parent Standard: Operational Risk & Trust Integrity Standard (ORTIS)

Version: 1.0

Status: Canonical · Open Module

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Compatibility: OOF Methodology OS · Operational Risk & Trust Integrity Standard (ORTIS) · Operational Evidence & Auditability Standard (OEAS) · Operational Escalation Integrity Standard (OESIS) · Operational Authority Integrity Standard (OAIS) · Runtime Integrity Standard (RIS) · Operational Decision Integrity Standard (ODIS) · INTEGROS® — Integrity Standard · Autonomous Systems · Multi-Agent Environments

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Trust Degradation Module (TDM) defines the structural conditions under which declining trustworthiness, trust-state deterioration, behavioral legitimacy degradation, operational confidence reduction, and consequence-bearing trust failures remain materially detectable, traceable, governable, and operationally valid across autonomous operational environments. TDM governs trust degradation. The module ensures that operational trust is treated as a dynamic condition rather than a permanent status.

A system satisfies TDM only if:

- trust degradation remains detectable
- trust-state transitions remain traceable
- declining trust remains governable

- behavioral legitimacy remains assessable
- trust reduction remains operationally actionable
- consequence-bearing trust failures remain accountable
- A system that continues granting trust-valid operational status despite material trust degradation does not satisfy TDM.

Module Operational Role

TDM defines the trust-degradation governance layer of ORTIS. Its role is to preserve governance-valid awareness whenever operational trust begins to deteriorate.

Module Operational Space

- TDM governs:
 - trust degradation
 - trust-state transitions
 - behavioral legitimacy decline
 - operational confidence reduction
 - trust deterioration monitoring
 - trust-governance continuity
 - consequence-bearing trust failure detection
 - trust-risk alignment preservation
- The module applies wherever operational trust may change during runtime operation.

Module Function

- The module applies wherever systems must preserve:
 - trust visibility
 - behavioral legitimacy assessment
 - trust-state awareness
 - governance-valid trust management
 - consequence-bearing trust governance
 - operational trust continuity
- Its function is to ensure that declining trust conditions are identified before they become operational failures.

Minimum Implementation Framework

1. Define the Trust Object

The organization must define which trust relationships require governance. This may include: agent trust tool trust system trust authority trust delegated trust operational trust consequence-bearing trust relationships

2. Define Trust Degradation Conditions

The system must define the conditions under which trust degradation becomes operationally significant. This includes: trust thresholds trust deterioration indicators legitimacy requirements operational confidence

conditions behavioral integrity conditions governance response triggers

3. Define Trust Degradation Detection Logic

The system must define how trust deterioration is identified. This may include: abnormal behavioral patterns repeated policy violations authority misuse execution anomalies evidence inconsistencies operational legitimacy decline governance-alignment degradation

4. Define Operational Response or Governance Logic

The system must define governance logic for trust degradation conditions. Governance response may include: trust-level reduction permission restriction

- authority limitation
- escalation activation
- governance review
- operational intervention
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

- The system must preserve reconstructable traceability of:
 - trust-state transitions
 - degradation events
 - governance decisions
 - intervention actions
 - authority adjustments
 - consequence-bearing outcomes
- A system must not remain trust-valid if material trust degradation remains undetected, unmanaged, or operationally ignored.

Use Case 1 — Autonomous Enterprise Agent

Network

Scenario

A distributed network of autonomous agents continuously performs operational tasks across enterprise systems and business environments.

Application

TDM governs trust-state transitions and identifies behavioral patterns indicating declining operational trustworthiness.

Result

The organization gains stronger trust visibility and reduced exposure to hidden trust degradation across autonomous operational environments.

Use Case 2 — Multi-Agent Coordination

Environment

Scenario

A multi-agent infrastructure continuously coordinates decisions, information exchange, and operational workflows across distributed runtime systems.

Application

TDM governs trust deterioration, operational confidence reduction, and governance-valid intervention when trust conditions weaken.

Result

The environment gains stronger operational legitimacy and reduced risk of trust-related coordination failures.

Canonical Closing Statement

Trust Degradation Module (TDM) defines the structural conditions under which declining trustworthiness, trust-state deterioration, behavioral legitimacy degradation, operational confidence reduction, and consequence-bearing trust failures remain materially detectable, traceable, governable, and operationally valid across autonomous operational environments. Operational trust is not valid merely because trust once existed. Trust governance becomes valid only when trust degradation remains detectable, traceable, governable, and capable of triggering legitimate operational response before consequence-bearing failures emerge.

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Canonical Source

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